

Primes and divisors

Previous Lecture

- *Integers*
- *Division, properties of divisibility*
- *The division algorithm*
- *Base b representation of numbers*
 - *Repeatedly divide by b and collect remainders*

$$n = bq_0 + r_0$$

$$q_0 = bq_1 + r_1$$

$$q_1 = bq_2 + r_2$$

...

$$q_k = bq_k + r_k$$



$$n = bq_0 + r_0$$

$$= b(bq_1 + r_1) + r_0$$

$$= b^2q_1 + br_1 + r_0$$

...

$$= b^k r_k + \dots + b^3 r_3 + b^2 r_2 + br_1 + r_0$$

Primes

- *Every integer n (except for 1 and -1) has at least 2 positive divisors, 1 and n itself (or $-n$).*
- *A positive number that does not have any other positive divisor is called **prime**.*
- *Examples:*
 - Small primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, ...*
 - Large primes: $2^{77232917} - 1$*
- *Mersenne numbers are numbers of the form $M_n = 2^n - 1$*
- *Greatest known Mersenne prime is $M_{77232917} = 2^{77232917} - 1$*
- *The next candidate is $M_{M_{61}} = 2^{2305843009213693951} - 1$*
- *A positive number that is not prime is called **composite***

Composite Numbers

- *Every composite number has a prime divisor.*
- *Proof. (by contradiction)*

*Let S be the set of all composite numbers that **do not** have a prime divisor*

Since $S \subseteq \mathbb{N}$, by the Well-Ordering Principle, it has a least element r .

As r is not prime, it has a divisor, i.e. $r = uv$ for some positive integers u and v with $u < r$ and $v < r$.

But $u \notin S$, so u has a prime divisor p .

Since $p \mid u$ and $u \mid r$, we conclude that $p \mid r$, a contradiction.

How many prime numbers are there?

- **Theorem (Euclid)**

There are infinitely many prime numbers.

- **Proof. (By contradiction)**

Suppose that $\{p_1, p_2, \dots, p_k\}$ is the set of all prime numbers, and let

$$a = p_1 p_2 \dots p_k + 1$$

Since a is greater than any member of the list, a is composite.

Hence a has a non-trivial prime divisor p_j

Since $p_j \mid a$ and, $p_j \mid p_1 p_2 \dots p_k$ we have $p_j \mid a - p_1 p_2 \dots p_k = 1$

So $p_j = 1$, contradiction

- *If n is a positive integer, then there are approximately $\frac{n}{\ln n}$ prime numbers not exceeding n*

Open Problems about Primes

● Goldbach's Conjecture

*Every positive **even** number can be represented as the sum of two prime numbers.*

For example: $4 = 2 + 2$, $8 = 5 + 3$, $42 = 37 + 5$

Goldbach's conjecture is known to be true for even numbers up to $4 \cdot 10^{18}$

● The Twin Prime Conjecture

Twin primes are primes that differ by 2, such as 3 and 5, 5 and 7, 11 and 13, etc.

The Twin Prime Conjecture asserts that there are infinitely many twin primes.

The record twin primes: $2,996,863,034,895 \cdot 2^{1,290,000} \pm 1$

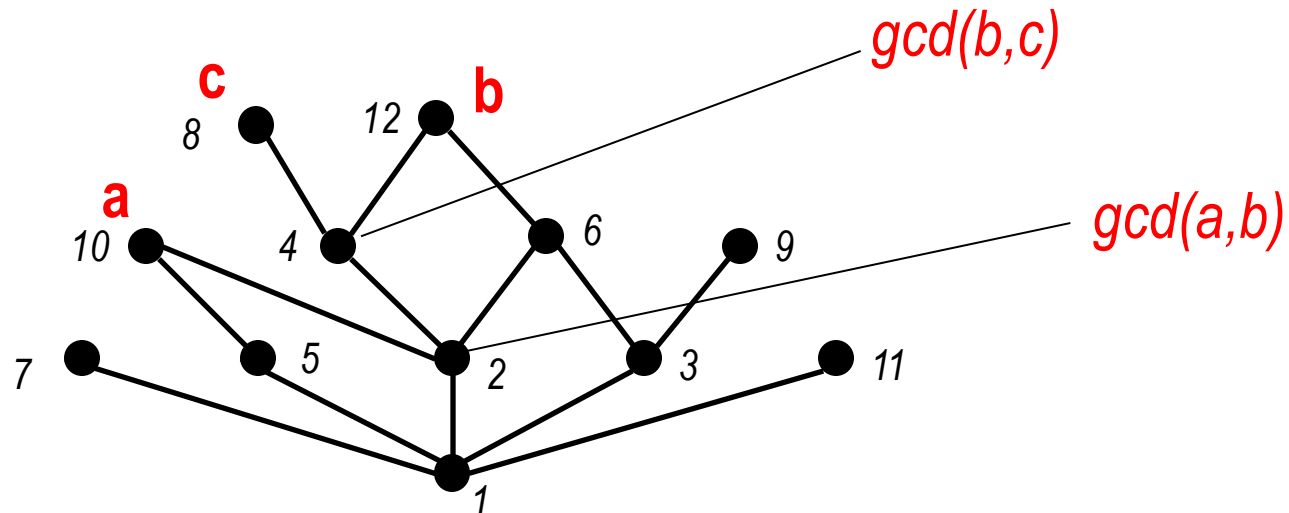
The Greatest Common Divisor

- For integers a and b , a positive integer c is said to be a **common divisor of a and b** if $c \mid a$ and $c \mid b$
- Let a, b be integers such that $a \neq 0$ or $b \neq 0$. Then a positive integer c is called the **greatest common divisor of a, b** if
 - $c \mid a$ and $c \mid b$ (that is c is a common divisor of a, b)
 - for any common divisor d of a and b , we have $d \mid c$
- What are the common divisors, and the greatest common divisor of 42 and 70?
- The greatest common divisor of a and b is denoted by $\gcd(a,b)$

The Greatest Common Divisor (cntd)

● Theorem

For any positive integers a and b , there is a unique positive integer c such that c is the greatest common divisor of a and b



The Greatest Common Divisor (cntd)

● Proof.

Given a, b , let $S = \{as + bt \mid s, t \in \mathbb{Z}, as + bt > 0\}$.

Since $S \neq \emptyset$, it has a least element $c = ax + by$.

We show that $c = \gcd(a, b)$

Bezout's identity

Step 1: $c \mid a$

If $c \nmid a$, by Euclidean division $a = qc + r$, where $0 < r < c$.

Then $r = a - qc = a - q(ax + by) = a(1 - qx) + b(-qy) \in S$

Hence we have a contradiction, so $c \mid a$. Similar for $c \mid b$

Step 2: Is it the greatest common divisor?

If $d \mid a$ and $d \mid b$, then $d \mid ax + by = c$.

Euclidean Algorithm: Small Example

- *To warm up, let us find the greatest common divisor of 287 and 91*
- *First let's divide 287 by 91:*
$$287 = 91 \cdot 3 + 14$$
- *Note that any common divisor of 91 and 14 also divides 287*
- *Also, $14 = 287 - 91 \cdot 3$, so any divisor of 287 and 91 also divides 14*
- *Thus, $\gcd(287, 91) = \gcd(91, 14)$!*
- *Now $91 = 14 \cdot 6 + 7$, so $\gcd(91, 14) = \gcd(14, 7)$!*
- *Since 7 divides 14,*
$$\gcd(14, 7) = 7 = \gcd(287, 91)$$

Euclidean Algorithm: Key Property

● Lemma.

*Let $a = bq + r$, where a , b , q , and r are integers.
Then $\gcd(a,b) = \gcd(b,r)$*

● Proof

Let d be a common divisor of a and b . Then d also divides $r = a - bq$. Thus, d is a common divisor of b and r .

Now, let d be a common divisor of b and r . Then d also divides $a = bq + r$.

*Therefore the pairs a,b and b,r have the same common divisors.
Hence, $\gcd(a,b) = \gcd(b,r)$.*

Euclidean Algorithm: The Algorithm

- Let a and b be positive integers with $a \geq b$. Set $r_0 = a$ and $r_1 = b$. Successively apply the division algorithm until the remainder is 0

$$r_0 = r_1 q_1 + r_2 \quad 0 < r_2 < r_1$$

$$r_1 = r_2 q_2 + r_3 \quad 0 < r_3 < r_2$$

$$\vdots$$

$$r_{k-2} = r_{k-1} q_{k-1} + r_k \quad 0 < r_k < r_{k-1}$$

$$r_{k-1} = r_k q_k$$

- Eventually, the remainder is zero, because the sequence of remainders $a = r_0 > r_1 > r_2 > \dots \geq 0$ cannot contain infinitely many elements.

- Furthermore, $\gcd(a, b) = \gcd(r_0, r_1) = \dots = \gcd(r_{k-2}, r_{k-1})$
 $= \gcd(r_{k-1}, r_k) = \gcd(r_k, 0) = r_k$

- Hence $\gcd(a, b)$ is the last nonzero remainder in the sequence

Greatest Common Divisor

● Theorem.

If a, b are integers and d is their greatest common divisor, then there are integers u, v such that $d = au + bv$.

● Proof.

We use the Euclidean algorithm and the notation $a = r_0, b = r_1, d = r_k$

We have

$$\begin{aligned}
 d &= r_k = r_{k-2} - r_{k-1}q_{k-1} \\
 &= r_{k-2} - (r_{k-3} - r_{k-2}q_{k-2})q_{k-1} \\
 &= (r_{k-4} - r_{k-3}q_{k-3}) - (r_{k-3} - (r_{k-4} - r_{k-3}q_{k-3})q_{k-2})q_{k-1} \\
 &\quad \vdots \\
 &= r_0u + r_1v = au + bv
 \end{aligned}$$

$$r_0 = r_1q_1 + r_2$$

$$\vdots$$

$$r_{k-3} = r_{k-2}q_{k-2} + r_{k-1}$$

$$r_{k-2} = r_{k-1}q_{k-1} + r_k$$

$$r_{k-1} = r_kq_k$$

Example

● Find $d = \gcd(821, 123)$ and u and v such that $d = 821u + 123v$

More Primes

● *Prime numbers have some very special properties with respect to division*

● **Properties of primes.**

(1) *If a, b are integers and p is prime such that $p \mid ab$ then $p \mid a$ or $p \mid b$.*

(2) *Let a_i be an integer for $1 \leq i \leq n$, and p is prime and $p \mid a_1 a_2 \dots a_n$ then $p \mid a_i$ for some $1 \leq i \leq n$*

The Fundamental Theorem of Arithmetic

● Theorem.

*Every integer $n > 1$ can be represented **uniquely as a product of primes**, up to the order of the primes.*

● Proof.

● Existence

By contradiction. Suppose that there is an $n > 1$ that cannot be represented as a product of primes, and let m be the smallest such number.

m is not prime, therefore $m = st$ for some s and t

But then s and t can be written as products of primes, because $s < m$ and $t < m$.

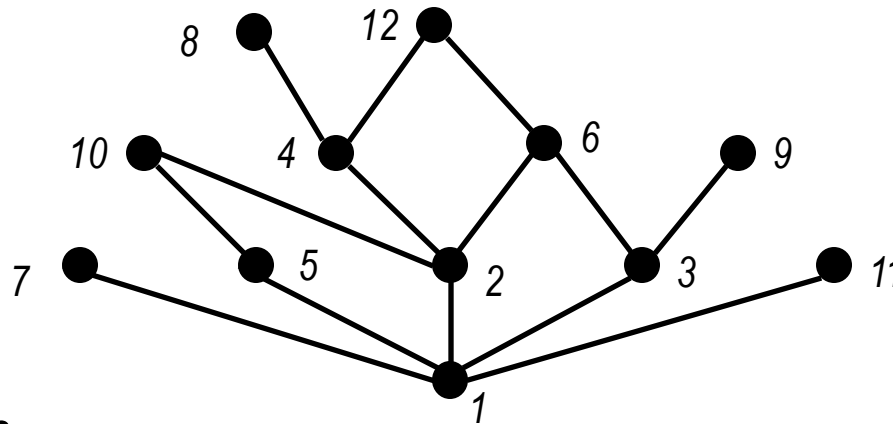
Therefore m is a product of primes

Example

● *Find the prime factorization of 4270*

Least Common Multiple

- A positive integer c is called a **common multiple** of integers a and b if $a \mid c$ and $b \mid c$
- The number c is called the **least common multiple** of a and b , denoted $\text{lcm}(a,b)$ if it is a common multiple and for any common multiple d we have $c \mid d$



● Theorem.

For any integers a and b , the least common multiple exists.

Least Common Multiple (cntd)

● Find $\text{lcm}(15,21)$.

● Theorem

For any integers a and b we have $ab = \text{lcm}(a,b) \cdot \text{gcd}(a,b)$

Relatively Prime

- Numbers a and b such that $\gcd(a,b) = 1$ are called **relatively prime**
- How many relatively prime numbers are there?
- **Euler's totient function** $\varphi(n)$ is the number of numbers k such that $0 < k < n$ and n and k are relatively prime.
- If p is prime then every $k < p$ is relatively prime with p . Hence, $\varphi(p) = p - 1$.
- **Lemma.**
If a and b are relatively prime then $\varphi(ab) = \varphi(a) \cdot \varphi(b)$
- **Corollary.**
If $n = p_1^{s_1} p_2^{s_2} \dots p_u^{s_u}$ is the prime factorization of n , then

$$\varphi(n) = n \cdot \left(1 - \frac{1}{p_1}\right)^{s_1} \left(1 - \frac{1}{p_2}\right)^{s_2} \dots \left(1 - \frac{1}{p_u}\right)^{s_u}$$

Practice

Exercises from the Book:

7th edition: 2ef, 3ef, 5, 22, 24b, 26, 31, 33cd (page 272 – 273)

8th edition: 2ef, 3ef, 5, 22, 24b, 26, 31, 33cd (page 288 – 289)